Leg Parts

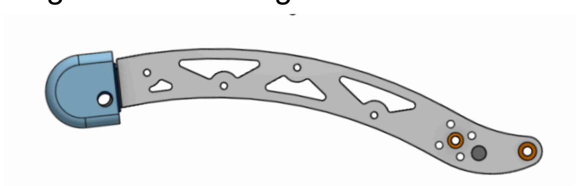
Design Goals:

- Lightweight
- Equal, 4-bar linkage
- High impact strength
- Accepts required sensors
- Affordability: Maintain low costs to maintain the budget
- Material Availability: accessible through standard consumer sources

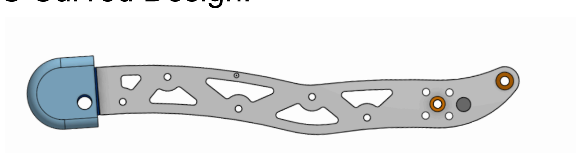
Calf:

Tube Design:

Weight: 0.16 lbs
Femur Design: Straight carbon fiber tube, off the shelf
Single Curved Design:



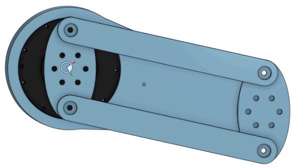
0.2536609 lb
Femur Design: 3d printed PETG frame with machined aluminum outer plates.
S Curved Design:



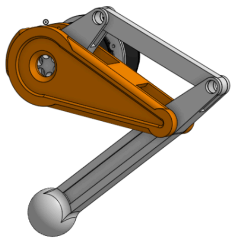
Weight: 0.288562 lbs
Femur Design: 3d printed PETG frame with machined aluminum outer plates.
Aligns with curve of thigh.

Leg Assembly:

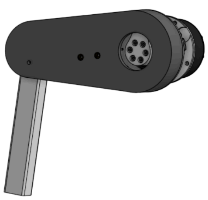
Iterations:



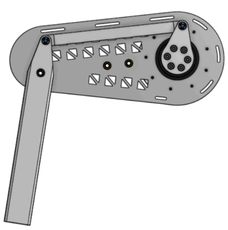
First complete thigh/quad assembly. Simple, but large and low range of motion.



Iteration introducing curved thigh/quad in pursuit of higher step clearance. 3D printed design.

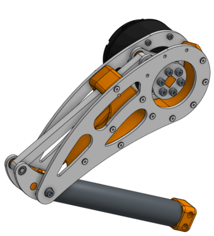


Thigh redesign with minimized linkage profile. 3D printed design.



Machined aluminum design featuring slots for 3D printed spacers and weight-saving cutouts

Curved design for increased step clearance, 3D printed thigh.



Final straight design featuring straight thigh, low profile linkage, machined aluminum plate with PETG inner profile.